



Chemistry: CYL1010

Textbooks

1. Silberberg, M., *Chemistry: The Molecular Nature of Matter and Change*, 6th Edition, McGraw Hill Education.
2. West, A.R., (2015), *Solid State Chemistry and Its Applications*, 2nd edition, John Wiley & Sons.
3. J. D. Lee, *Concise Inorganic Chemistry*, 5th Edition.
4. Atkins, *Physical Chemistry*, 11th Edition.

Reference Books

1. McMurry, J. E. & Fay, R. C. *Chemistry*, 5th Edition, Pearson.
2. Jonathan Clayden, Nick Greeves, Stuart Warren, *Organic Chemistry*, 2nd Edition.
3. F. Albert Cotton, Geoffrey Wilkinson, *Basic Inorganic Chemistry*, Wiley Student Edition.

Self Learning Material

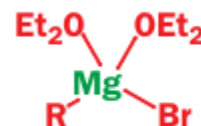
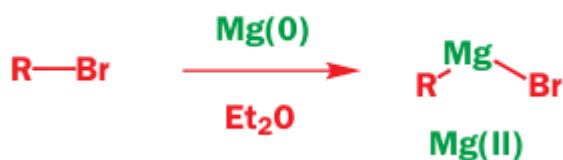
1. <https://nptel.ac.in/downloads/122101001/>

Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

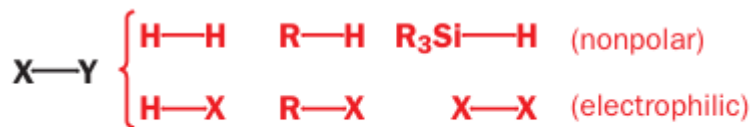
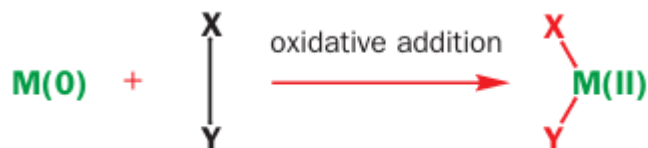
Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

Oxidative addition inserts metal atoms into single bonds

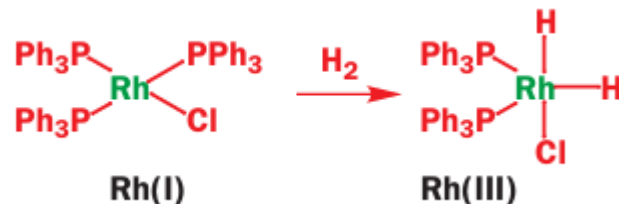
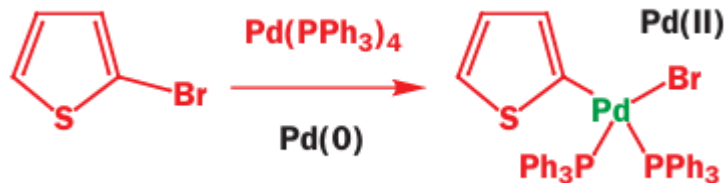
Potential ligands that do not have a lone pair or filled π type orbital are still able to interact with transition metal complexes but only by breaking a σ bond. This is the first step in a wide variety of processes and is described as oxidative addition because the formal oxidation state of the transition metal is raised by two, for example, $M(0)$ to $M(II)$, in the process. This is the result of having two extra ligands bearing a formal negative charge.



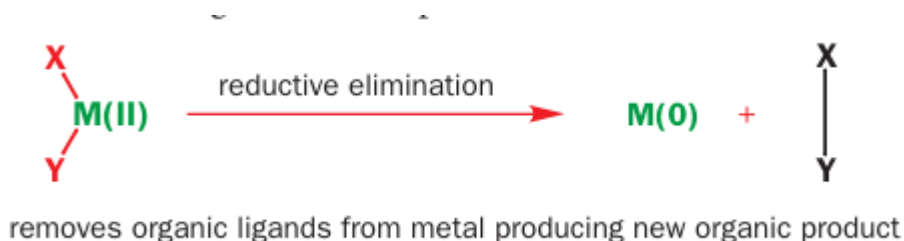
correct Grignard reagent structure has some (probably 2) ether ligands



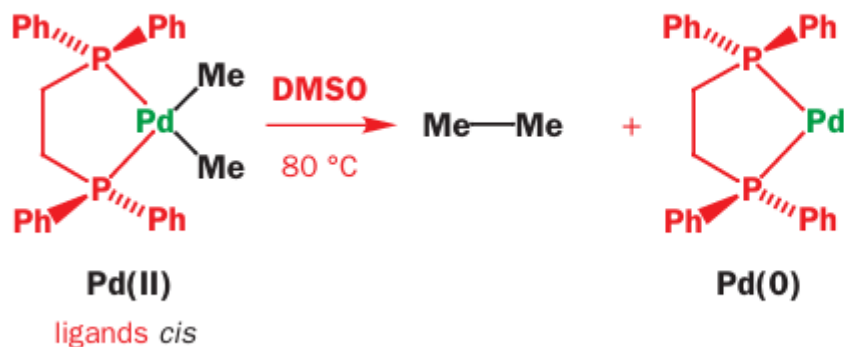
introduces new organic ligands on to metal



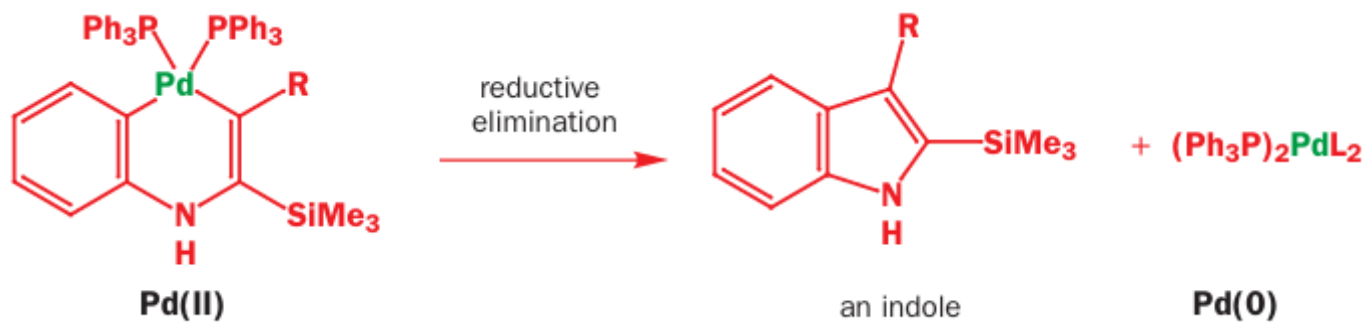
Reductive elimination



metal from a reaction sequence leaving a neutral organic product.

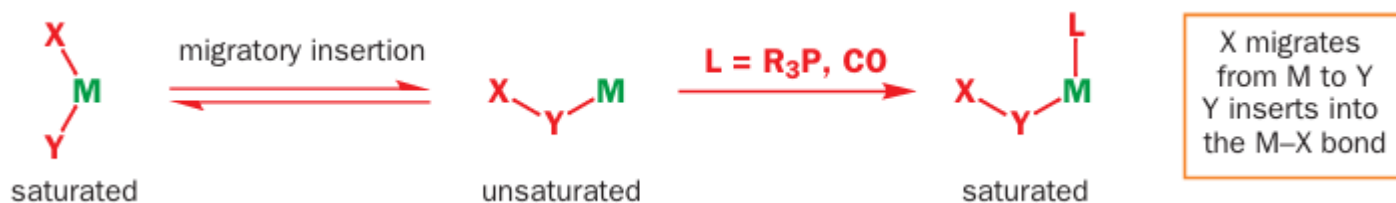


The ligands to be eliminated must be *cis* to one another for reductive elimination to occur. This is because the process is concerted. Reductive elimination is one of the most important methods for the removal of a transition metal from a reaction sequence leaving a neutral organic product.



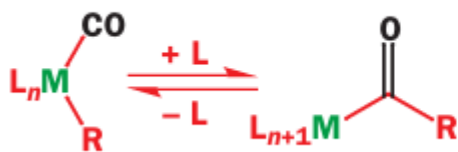
Migratory insertion builds ligand structure

- This process involves migration of one of the ligands from the metal to the other ligand and insertion of one of the ligands into the other metal–ligand bond and is known as migratory insertion.
- The insertion process is reversible and, as the metal effectively loses a ligand in the process, the overall insertion may be driven by the addition of extra external ligands (L) to produce a coordinatively saturated complex.
- As with reductive elimination, a *cis* arrangement of the ligands is required and the migrating group (X) retains its stereo chemistry (if any) during the migration.

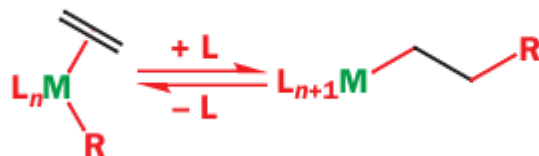


- Migratory insertion is the principal way of building up the chain of a ligand before elimination.
- The group to be inserted must be unsaturated in order to accommodate the additional bonds and common examples include carbon monoxide, alkenes, and alkynes producing metal–acyl, metal–alkyl, and metal–alkenyl complexes, respectively.
- In each case the insertion is driven by additional external ligands, which may be an increased pressure of carbon monoxide in the case of carbonylation or simply excess phosphine for alkene and alkyne insertions.

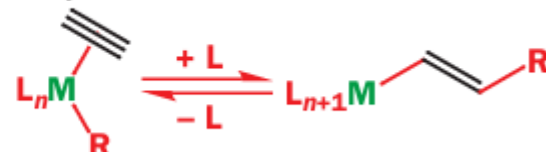
carbonylation



carbometallation or hydrometallation, R = H



alkyne insertion

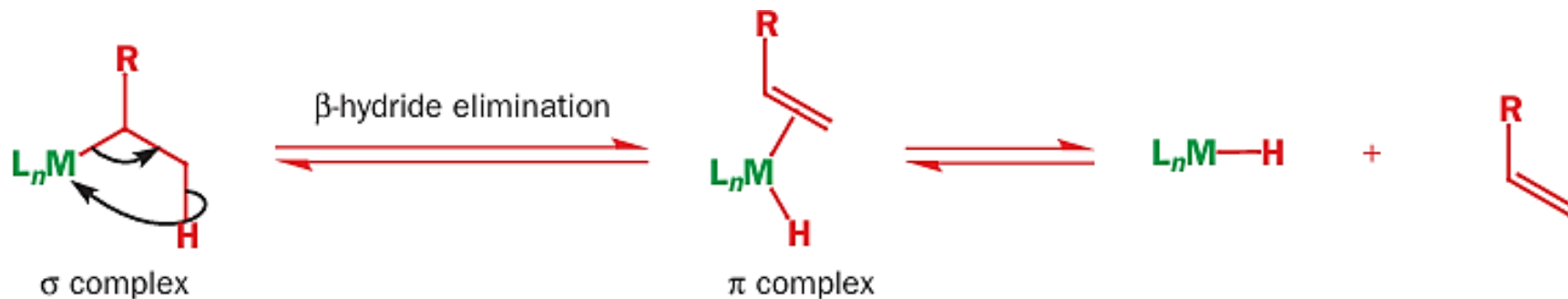


A good example of the carbonylation process is the reaction of the tetracarbonyl ferrate dianion $[\text{Fe}(\text{CO})_4]^{2-}$ with alkyl halides. This reagent is made by dissolving metal reduction of the 18 electron $\text{Fe}(0)$ compound $\text{Fe}(\text{CO})_5$. Addition of two electrons would give an unstable 20-electron species but the loss of one of the ligands with its two electrons restores the stable 18-electron structure.



Insertion reactions are reversible

- The reverse of hydrometallation involves the elimination of a hydride from the adjacent carbon of a metal alkyl to form an alkene complex. This process is known as β -hydride elimination or simply β elimination.
- It requires a vacant site on the metal as the number of ligands increases in the process and so is favored by a shortage of ligands as in 16-electron complexes. The metal and the hydride must be syn to each other on the carbon chain for the elimination to be possible. The product is an alkene complex that can lose the neutral alkene simply by ligand exchange.

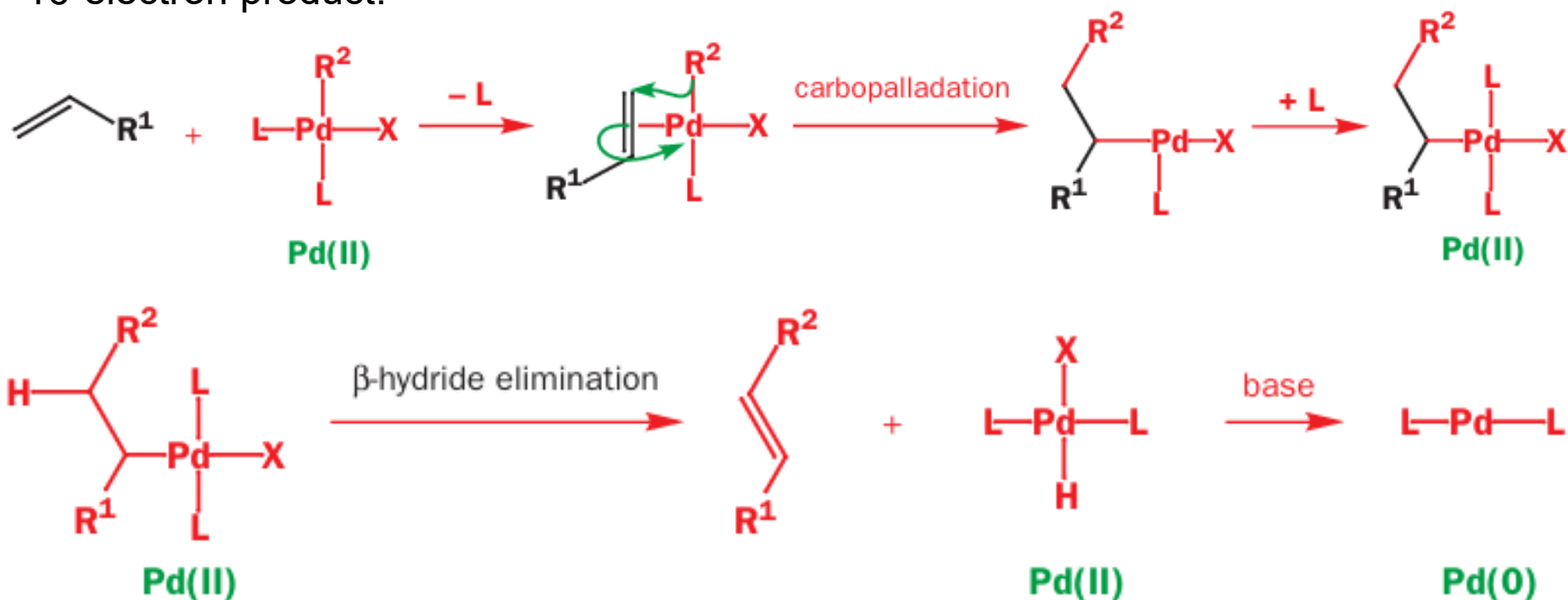


Palladium(0) is most widely used in homogeneous catalysis

Palladium chemistry is dominated by two oxidation states. The lower, palladium(0), present in tetrakis(triphenylphosphine)palladium, for example, is electron-rich, and will undergo oxidative addition with suitable substrates such as halides and triflates ($\text{TfO}^- = \text{CF}_3\text{SO}_2\text{O}^-$), resulting in a palladium(II) complex. Oxidative addition is thought to occur on the coordinatively unsaturated 14-electron species, formed by ligand dissociation in solution



The resulting σ alkyl bond in such complexes is very reactive, especially towards carbon–carbon π -bonds. Thus, an alkene in the reacting system will lead to coordination followed by migratory insertion into the palladium–carbon σ bond. This process is like hydrometallation and is called carbopalladation as carbon and palladium are attached to the ends of the alkene system. There is no change in oxidation state during this process, although the ligands (often phosphines) must dissociate to allow coordination of the alkene and associate to provide a stable final 16-electron product.



The Heck Reaction



R¹: aryl, heteroaryl, vinyl, or any alkyl group without β Hs

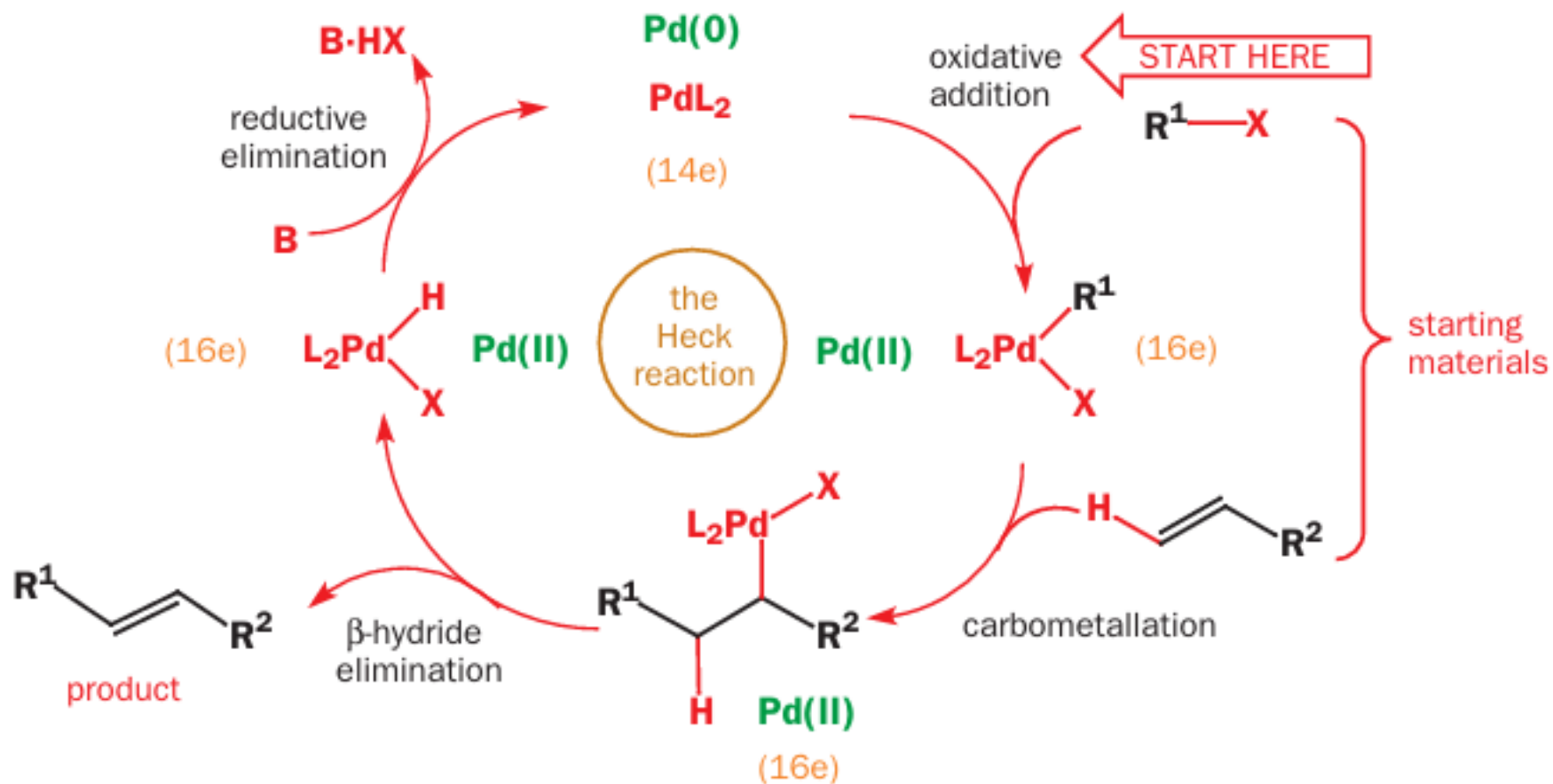
X: Halide (Br or I) or triflate (OSO₂CF₃)

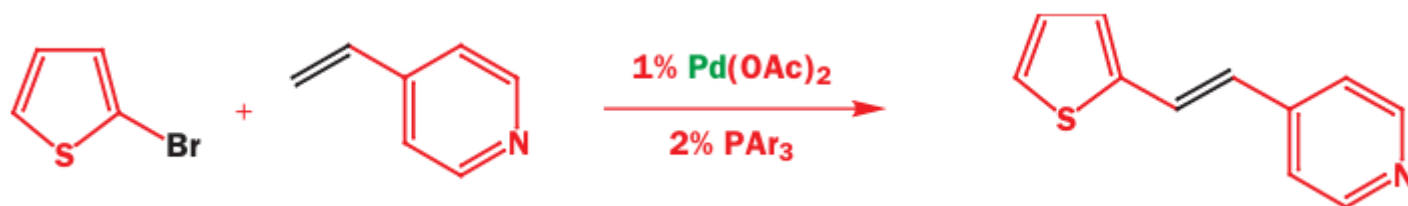
Alkene: mono- or disubstituted and can be electron-rich,-poor, or -neutral.

Base: Et₃N, NaOAc, or aqueous Na₂CO₃

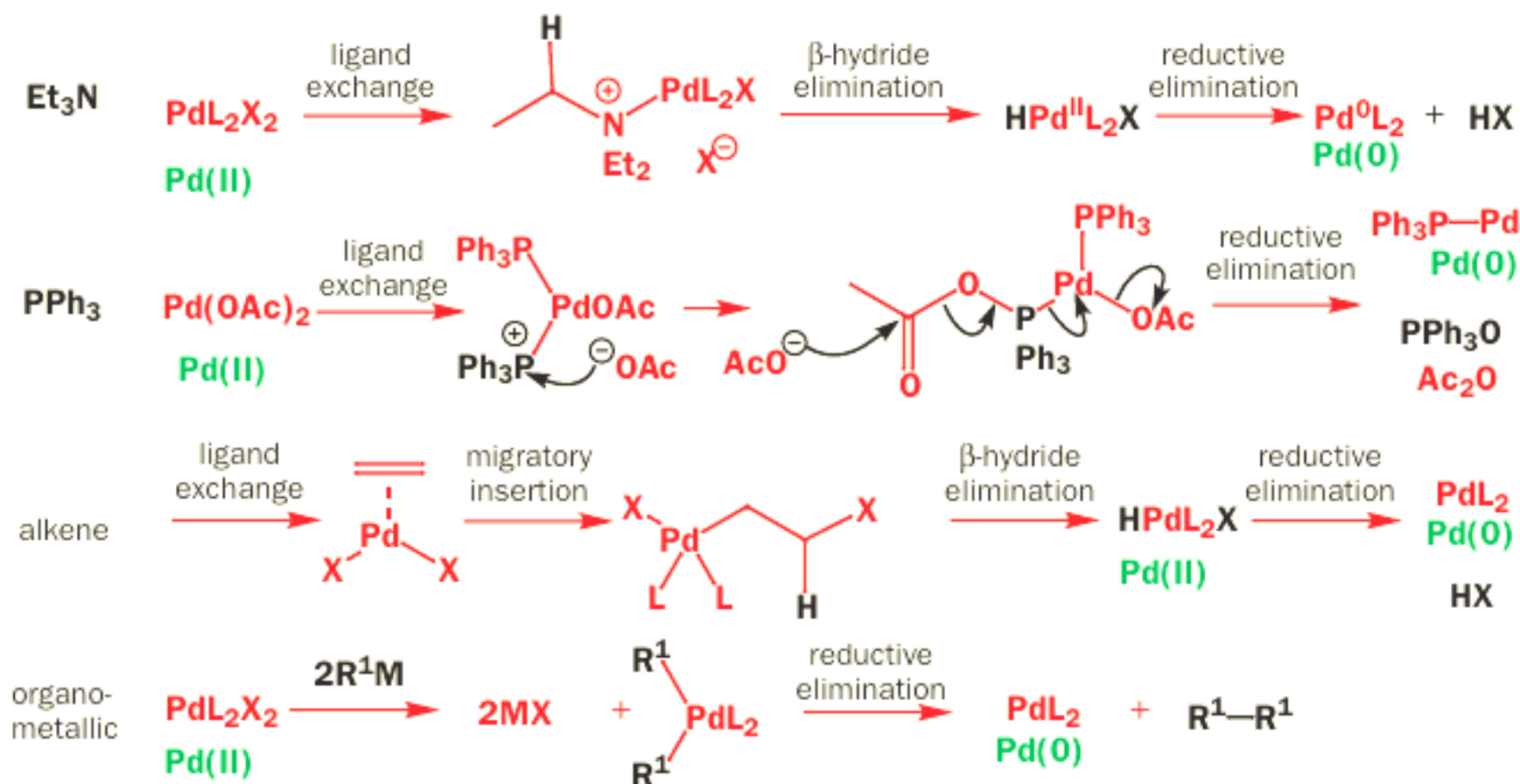
- Oxidative addition of the halide
- Insertion of the olefin
- Elimination of the product by a β-hydride elimination process
- A base then regenerates the palladium(0) catalyst

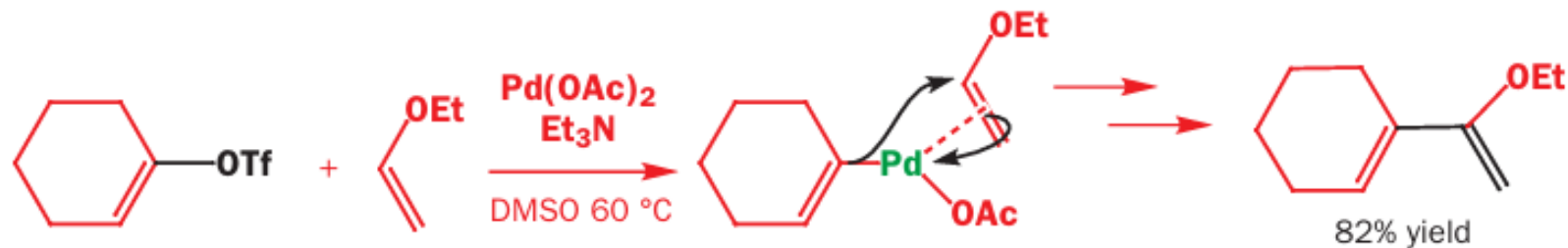
The Heck Reaction



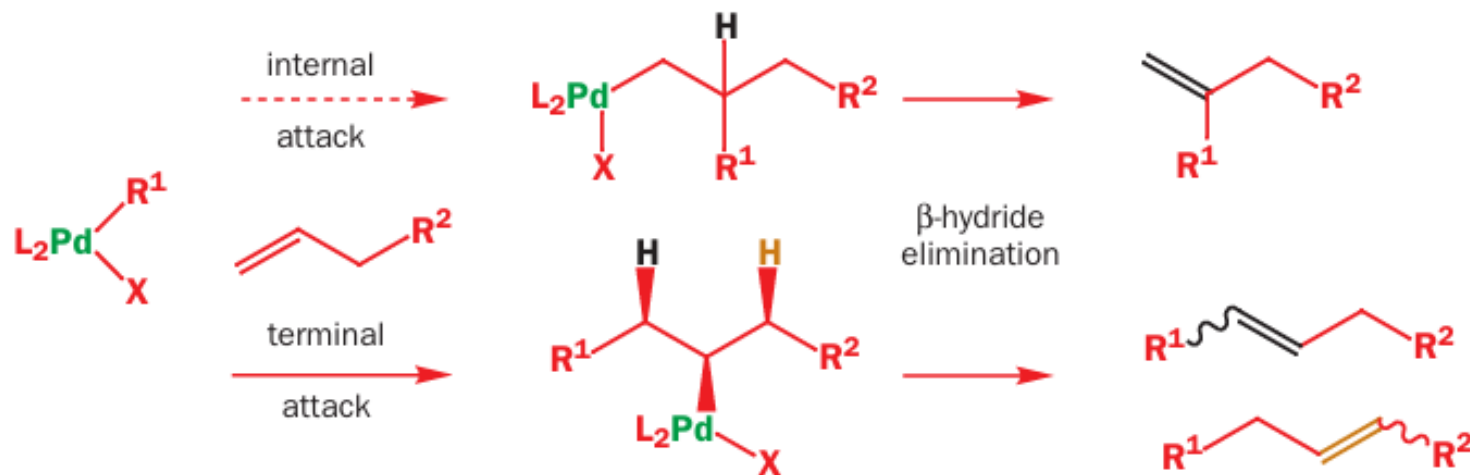


In situ formation of palladium(0) by reduction of Pd(II)



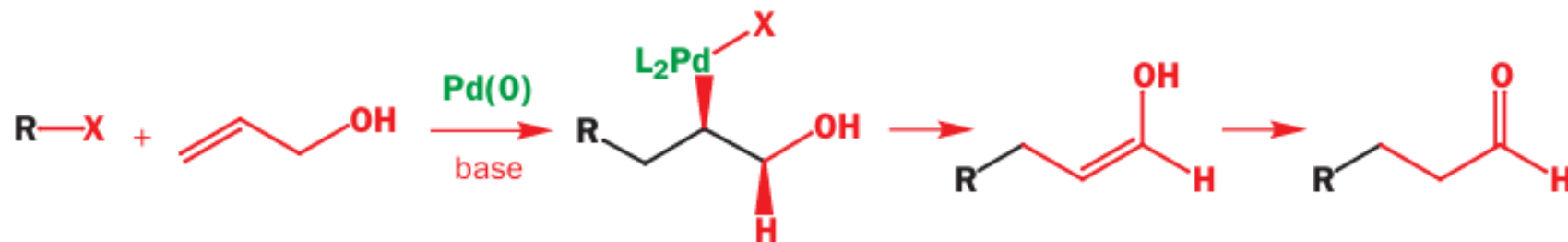


Electron-donating groups such as ethers lead to attack at the end of the alkene substituted by oxygen to produce in this case the 1,1-disubstituted product. These reactions must be dominated by the interaction of the filled π orbital of the alkene with an empty d orbital on Pd. This is an example of a Heck reaction working in the absence of a phosphine ligand

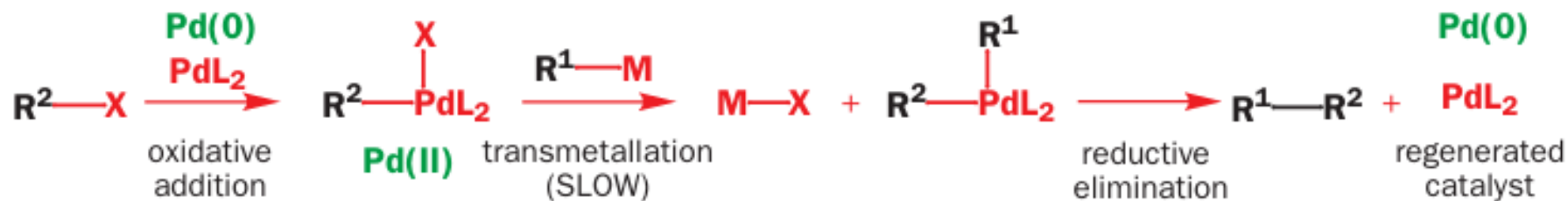
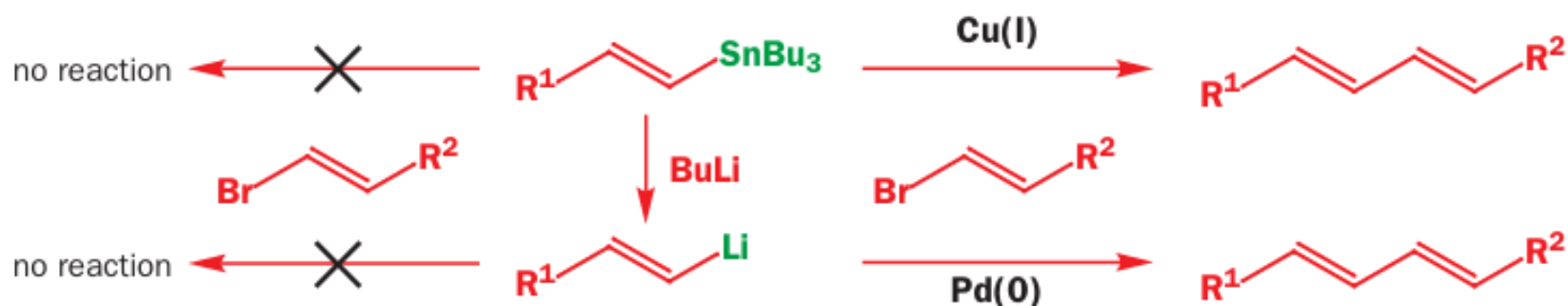
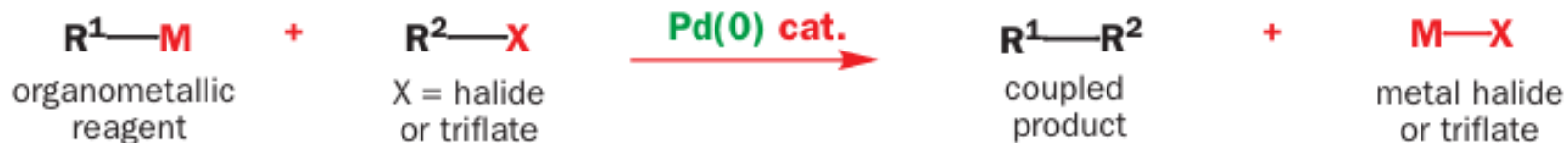


In the β -hydride elimination step, the palladium and hydrogen must be coplanar for reaction to take place, as this is a syn-elimination process. For steric reasons, the R group will tend to eclipse the smallest group on the adjacent carbon as elimination occurs, leading predominantly to a trans double bond in the product

Where there is a choice as to which hydride can be lost to form the alkene, the stability of the possible product alkenes often governs the outcome as the β -hydride elimination is reversible. The reaction of allylic alcohols is particularly important as the more stable of the two alkenes is the enol and a carbonyl compound is formed.



Cross-coupling of organometallics and halides





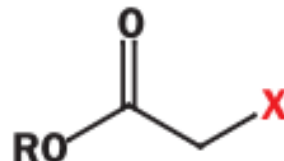
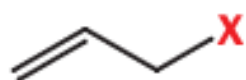
R^1-M $\text{R}^1 =$ almost anything including examples with β H

$\text{M} = \text{MgX}, \text{ZnX}, \text{Cu}, \text{SnR}_3, \text{SiR}_3/\text{TASF}, \text{ZrCp}_2\text{Cl}, \text{AlMe}_2, \text{B(OR)}_2$

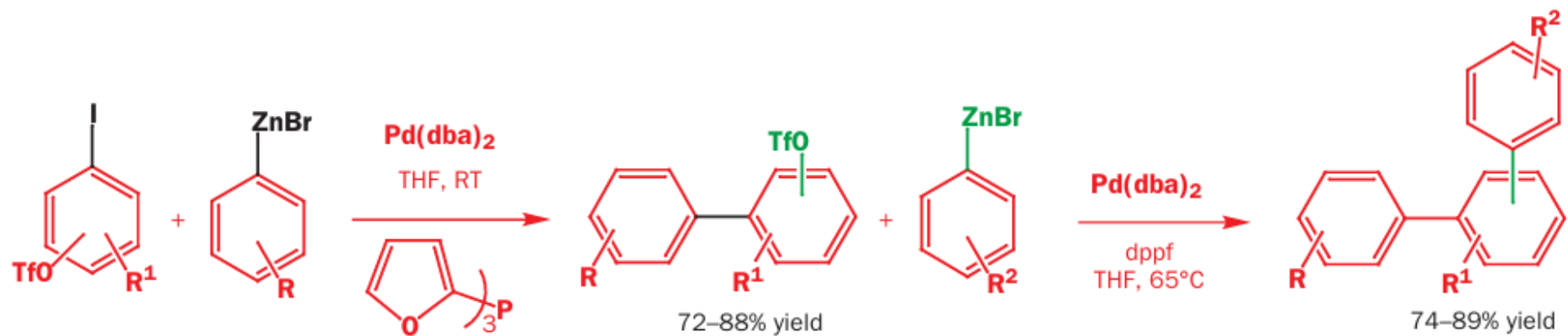
R^2-X R^2 must not have β Hs that can eliminate

$\text{X} = \text{I}, \text{Br}, (\text{Cl}), \text{OTf}, \text{OPO(OR)}_2$

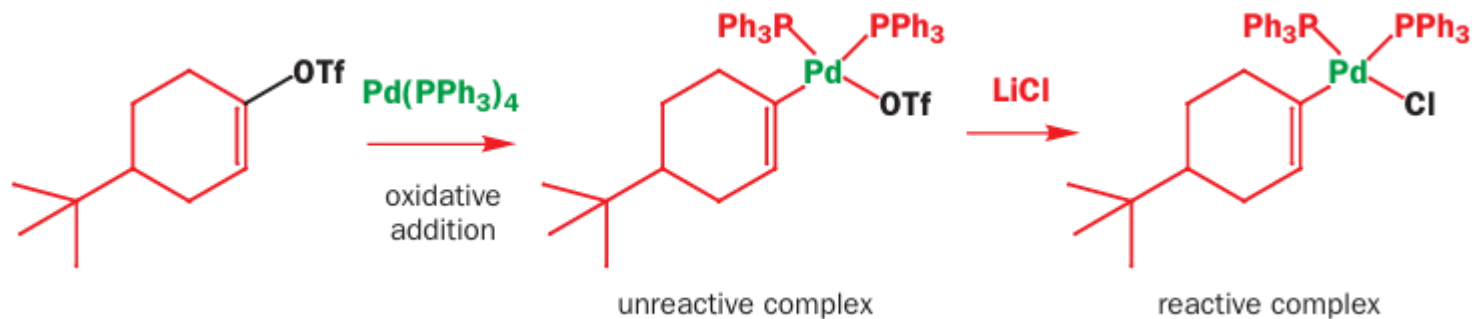
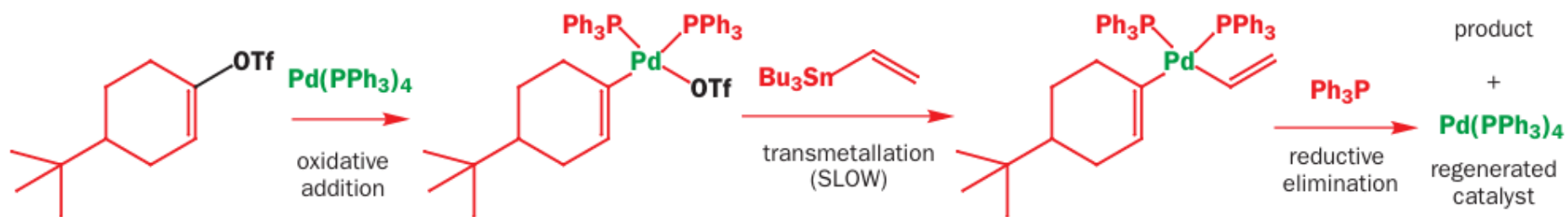
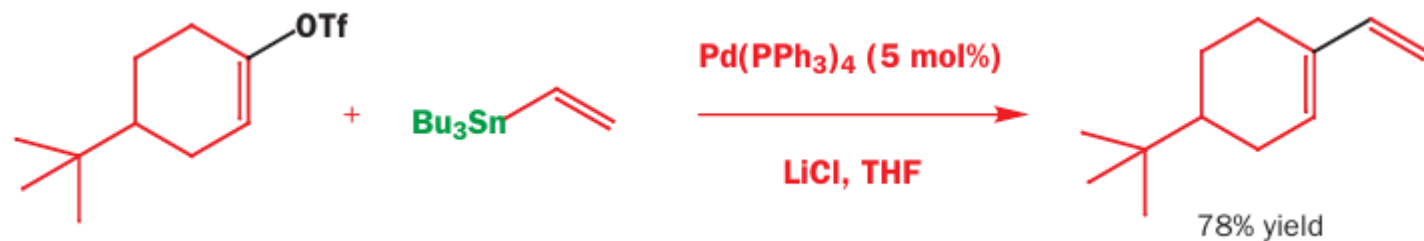
$\text{Ar}-\text{X}$

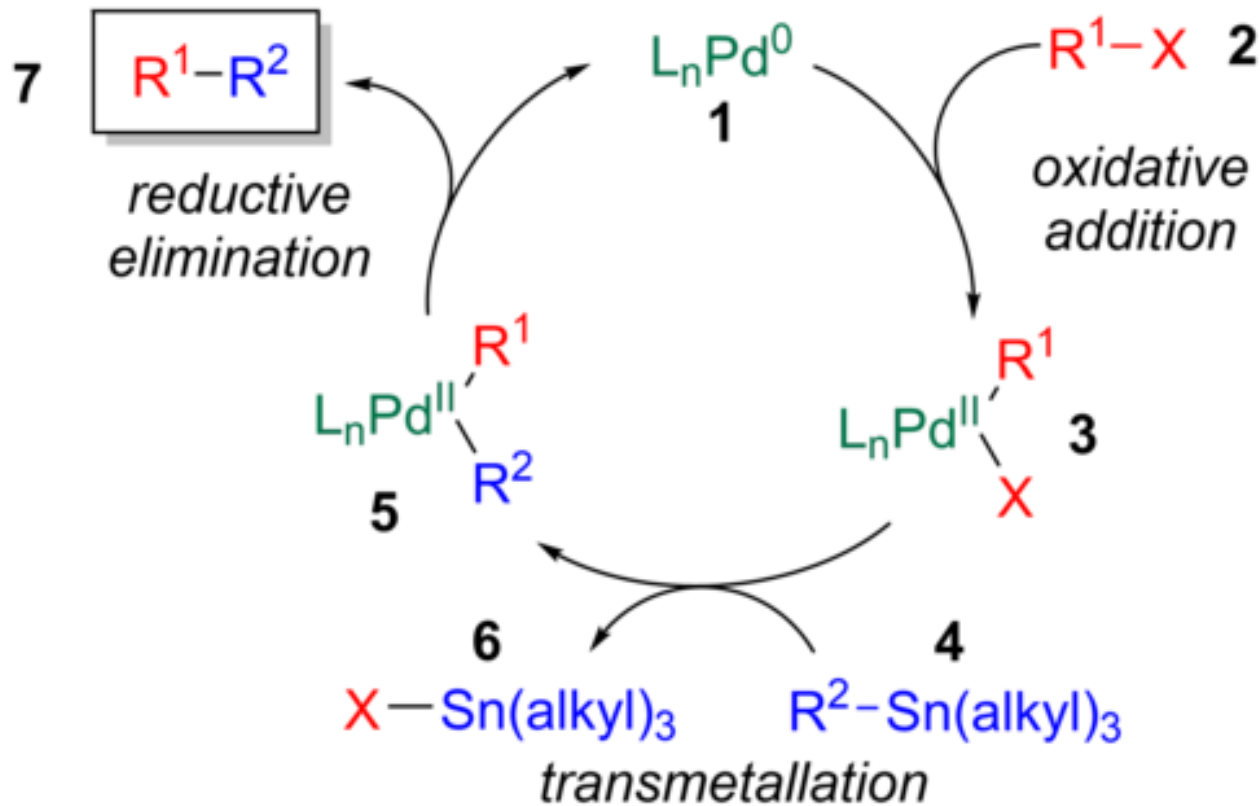


$\text{R}_\text{F} =$ per-fluoroalkyl



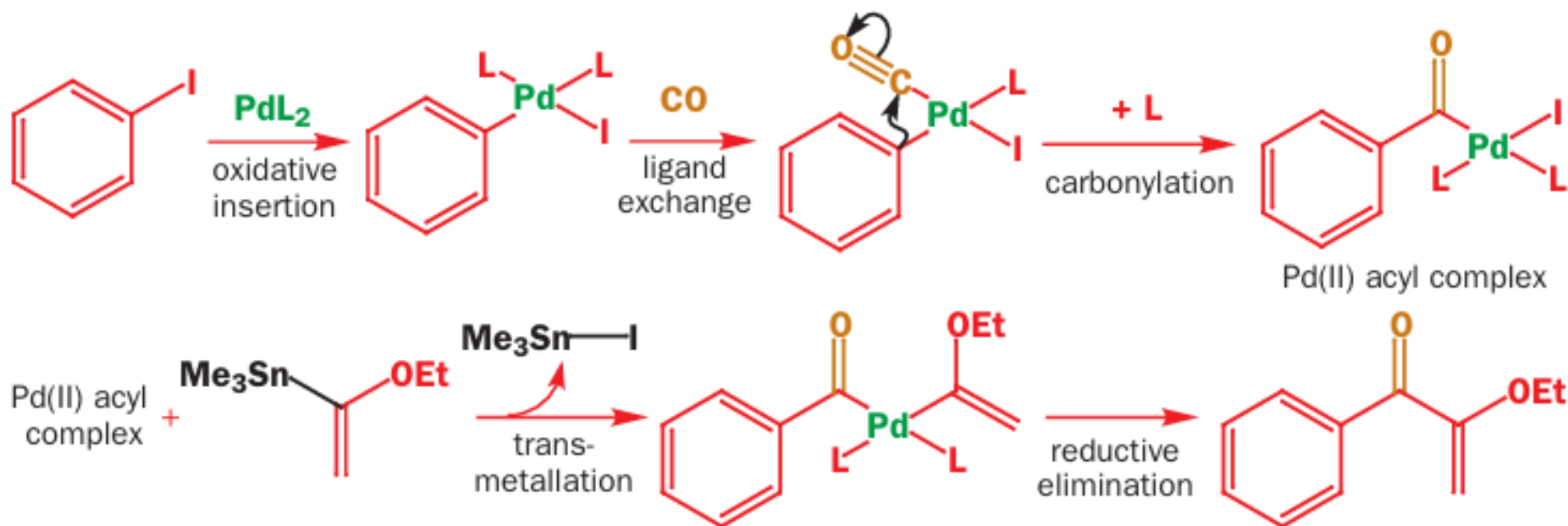
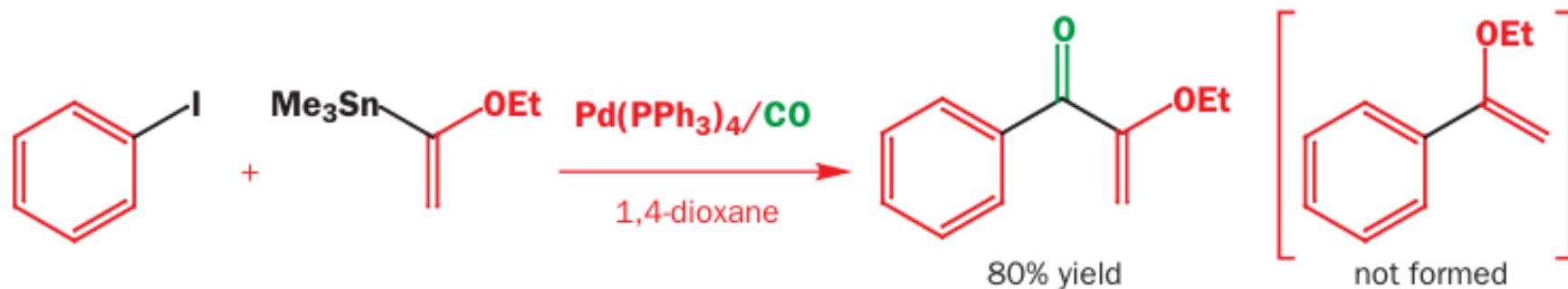
The Stille coupling



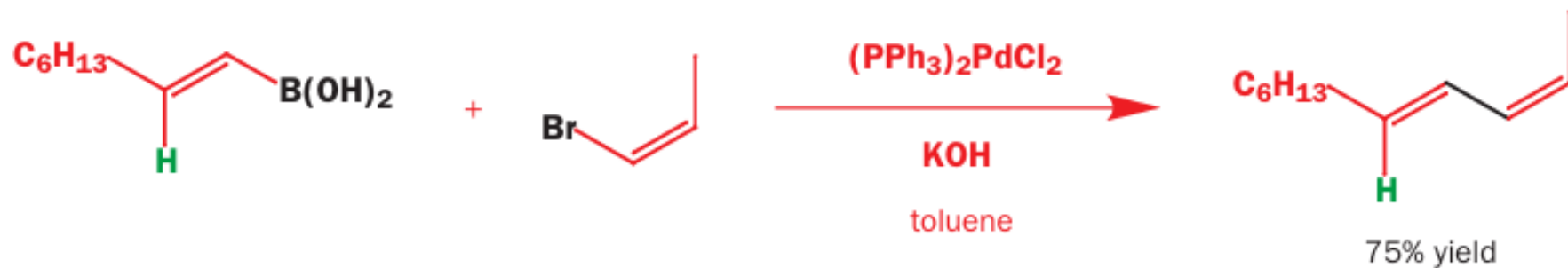
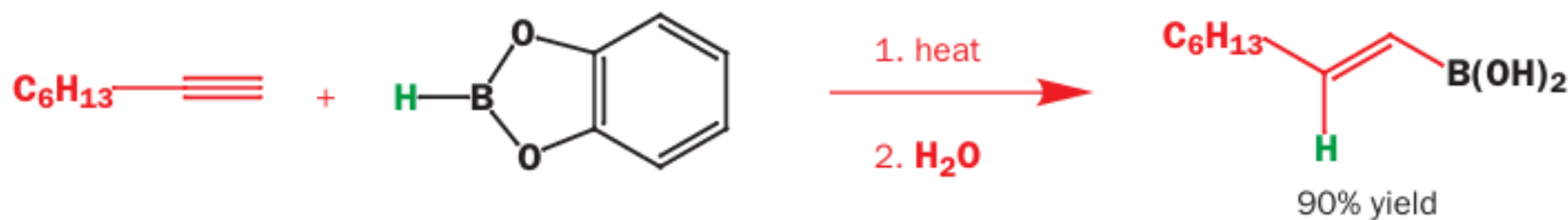
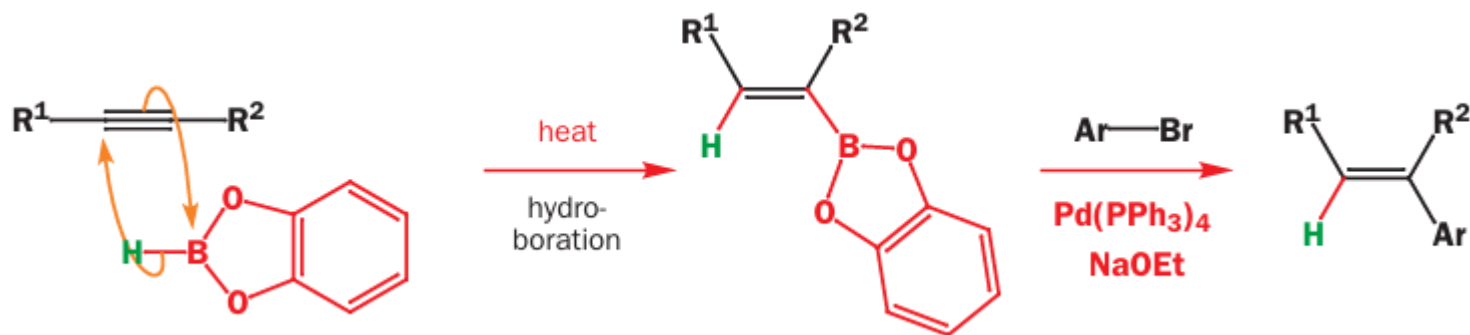


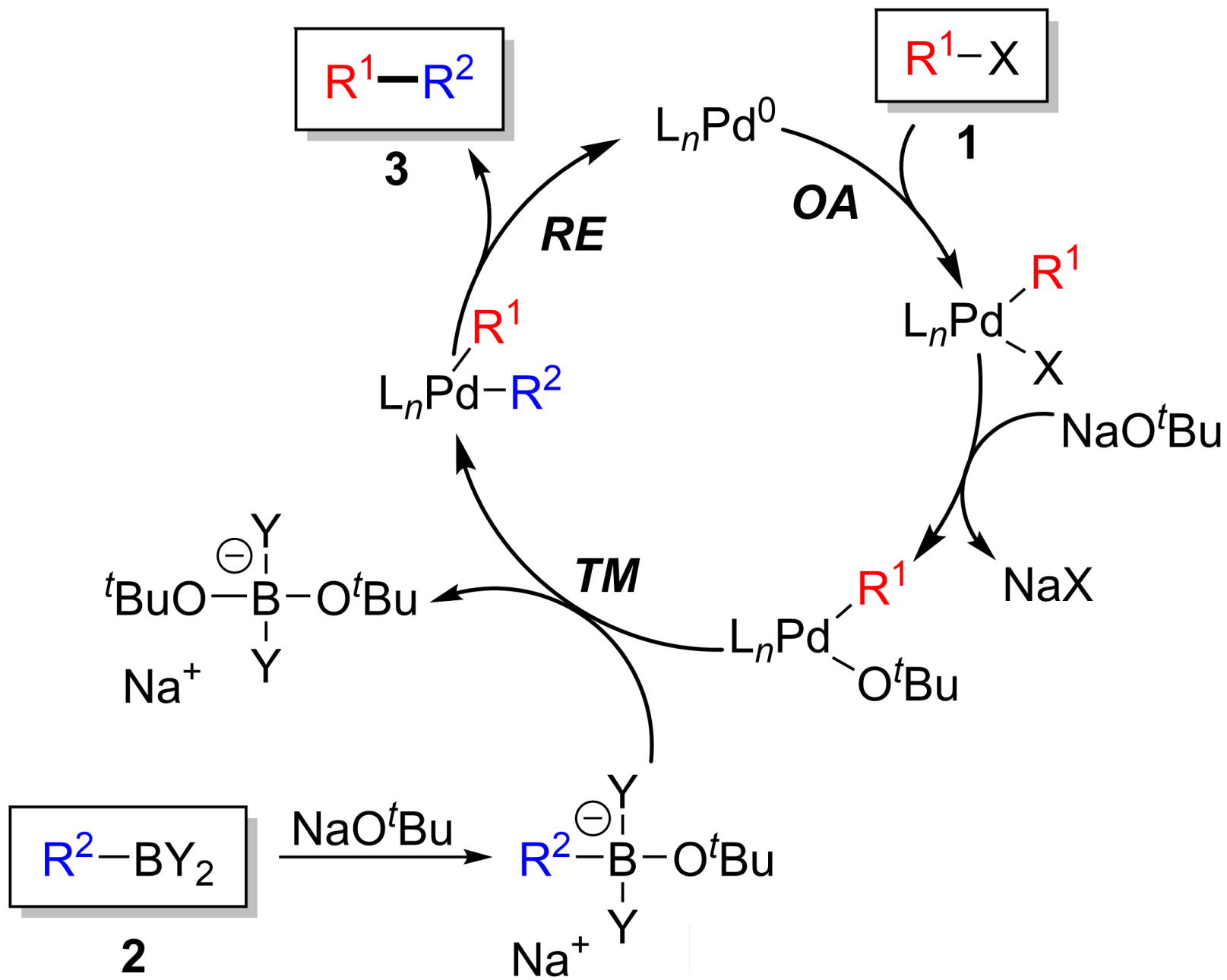
R^1, R^2 = allyl, alkenyl, aryl; X = Cl, Br, I, OTf, etc.

L = phosphine; alkyl = Me, Bu

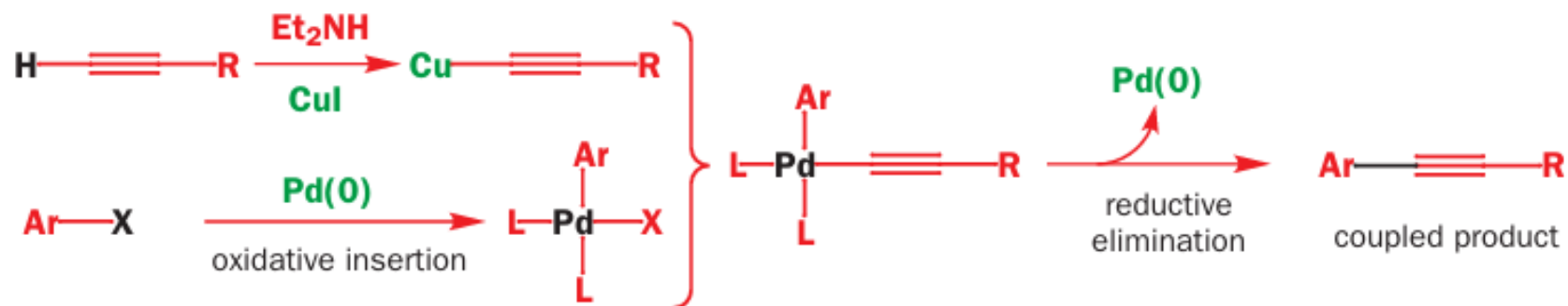
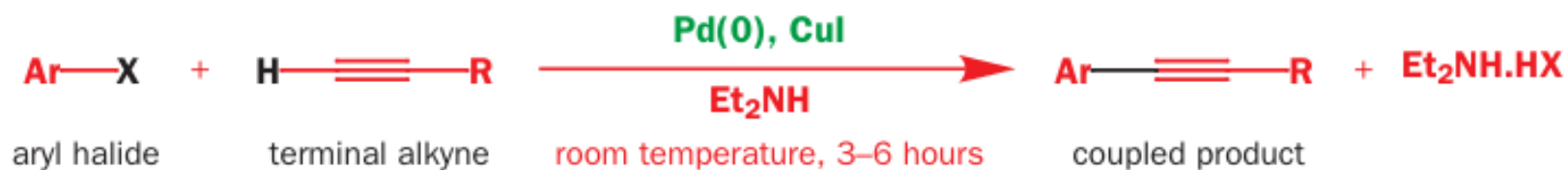


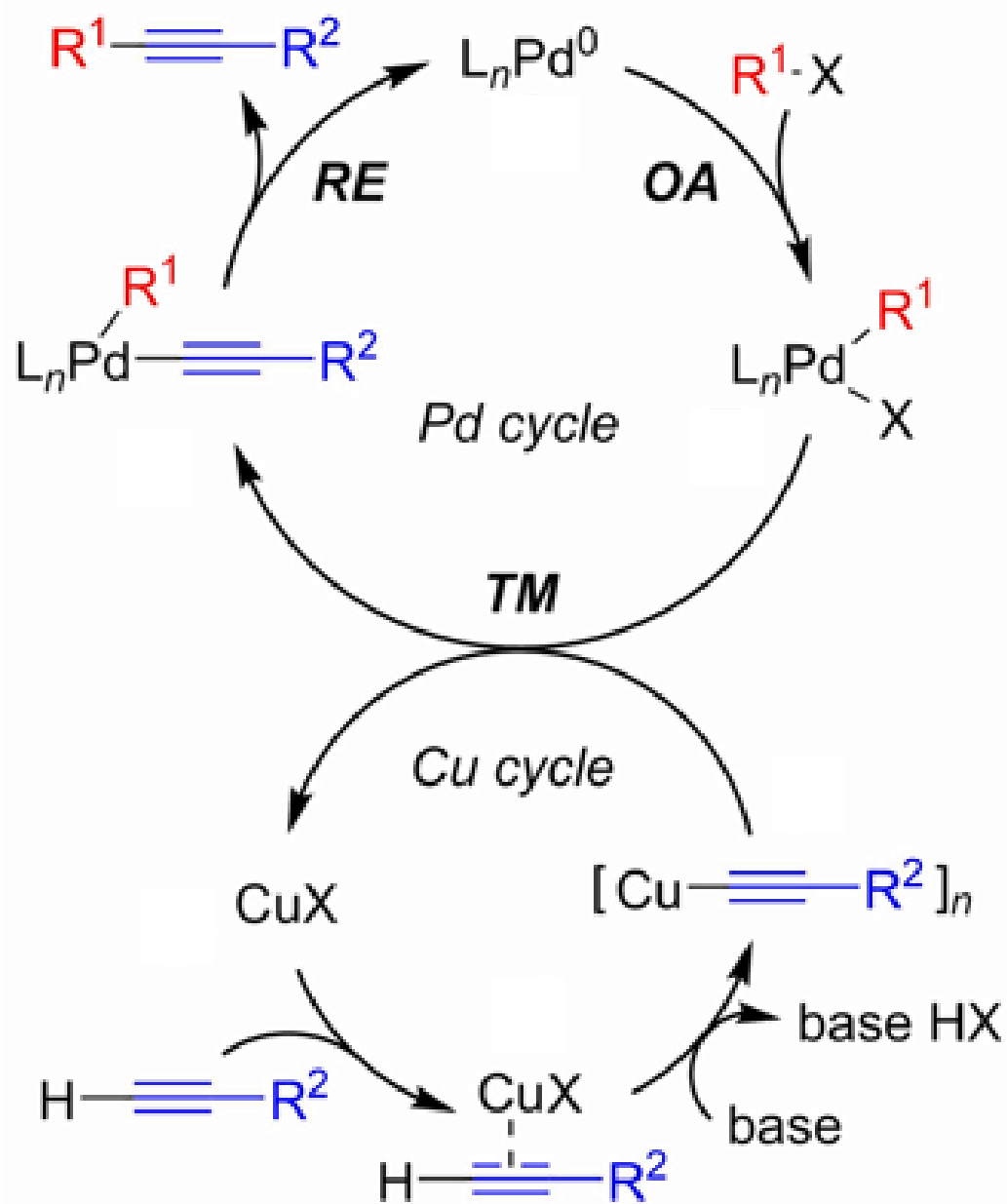
The Suzuki coupling

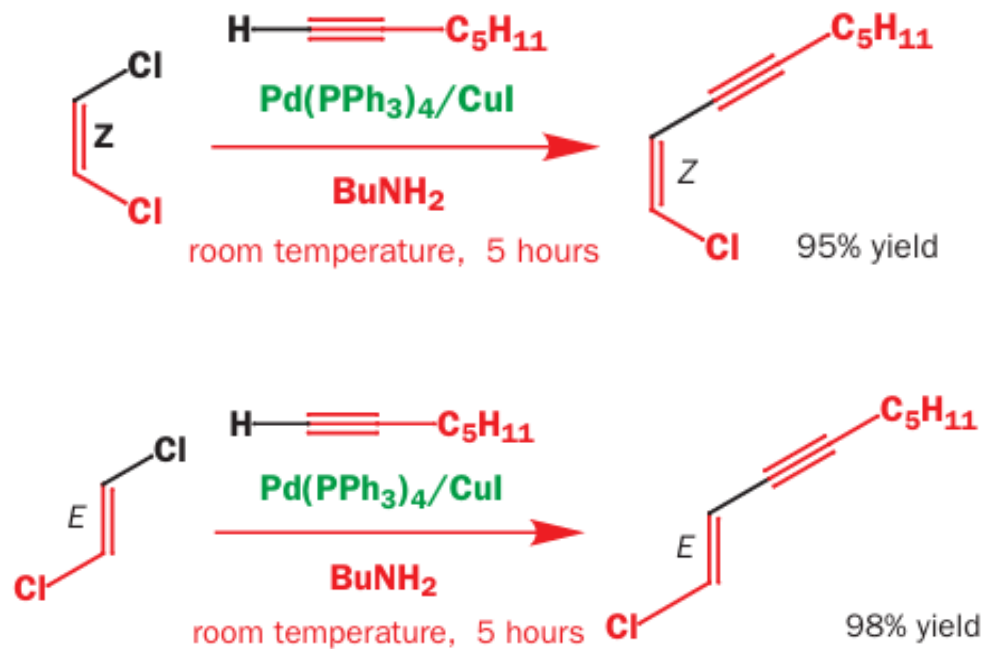




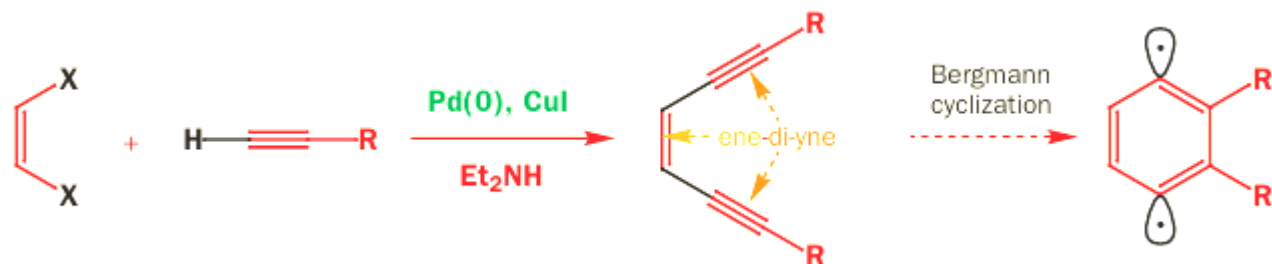
The Sonogashira coupling



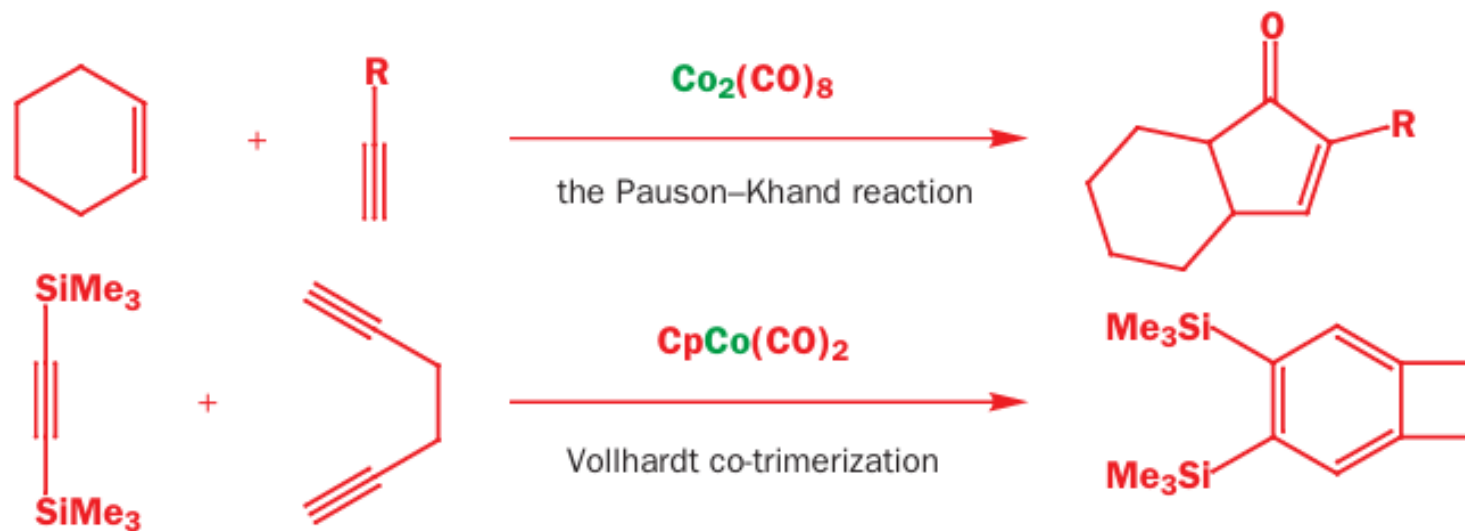




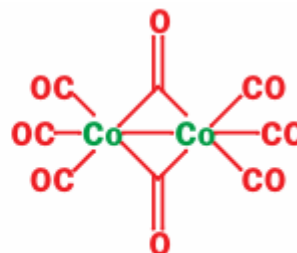
Bergmann cyclization

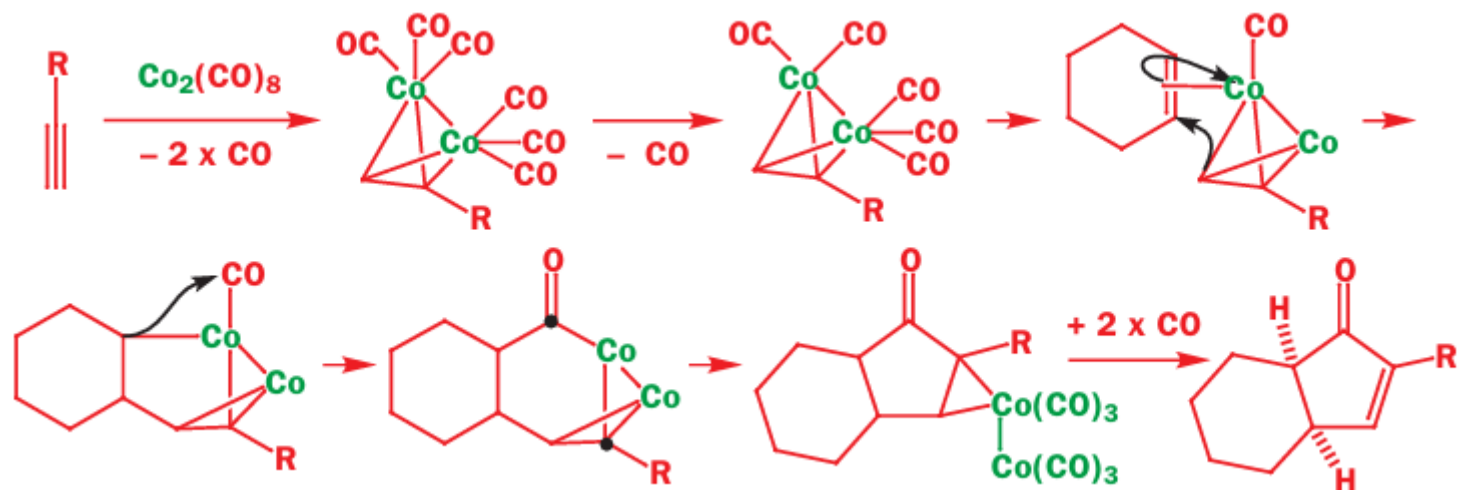


Other transition metals:cobalt



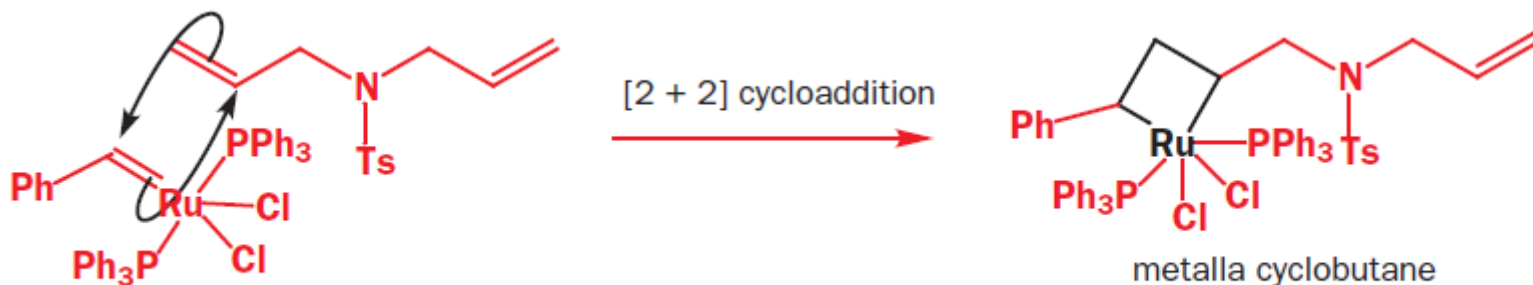
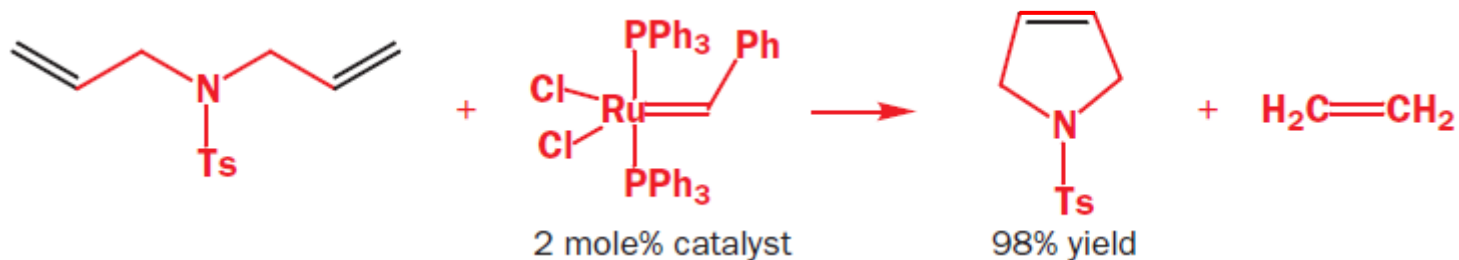
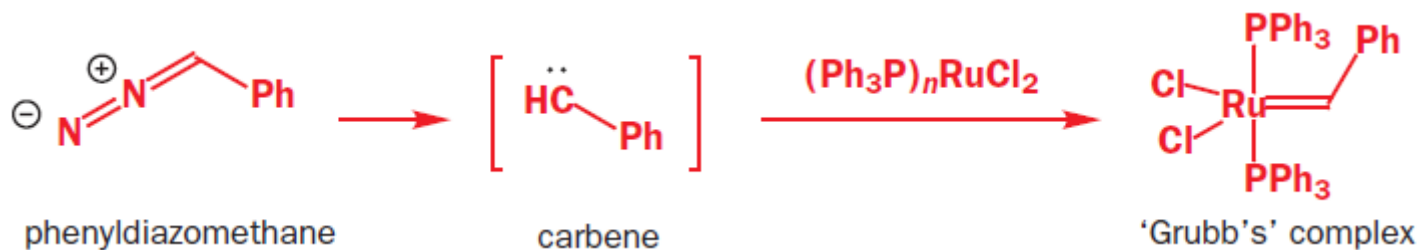
18-electron complex of $\text{Co}(0)$

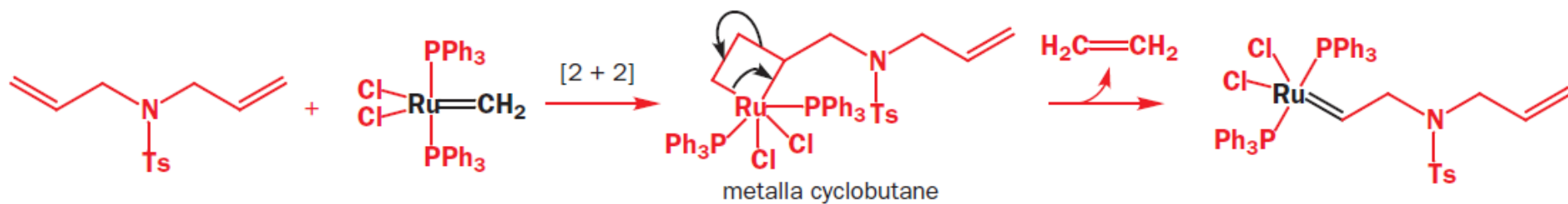
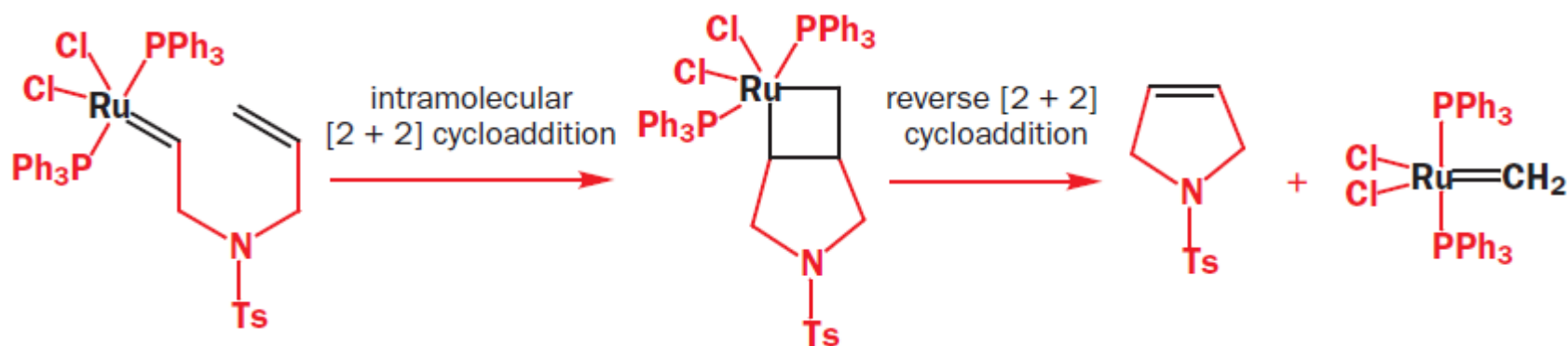
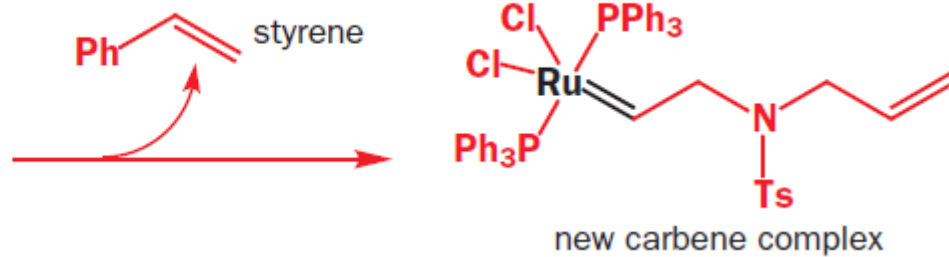
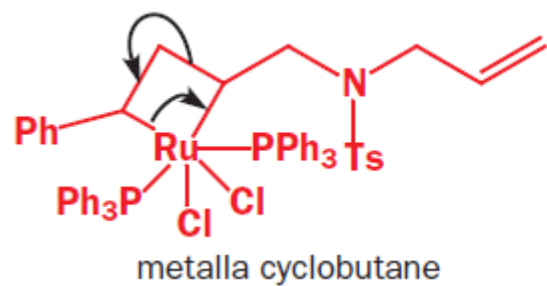


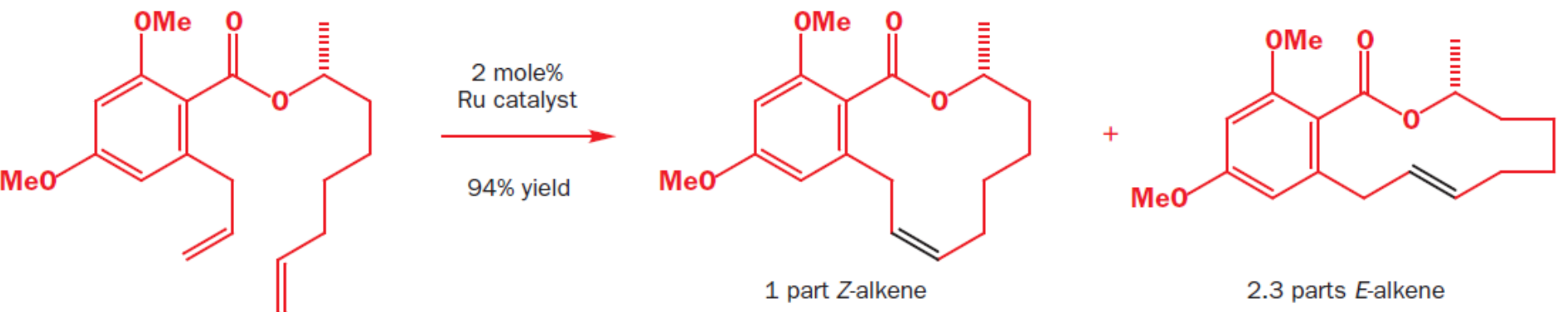
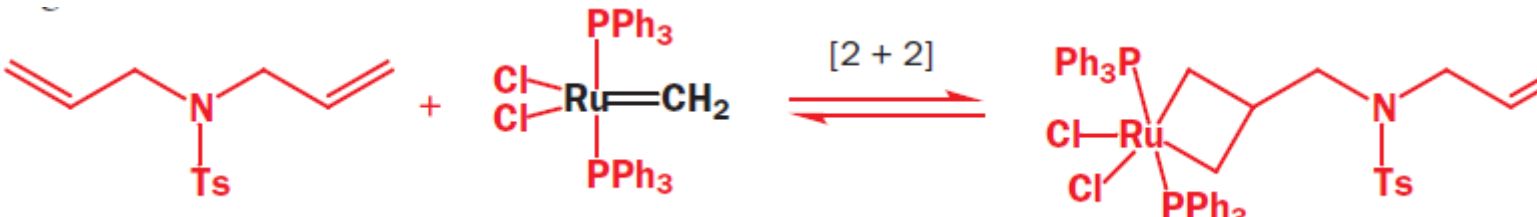


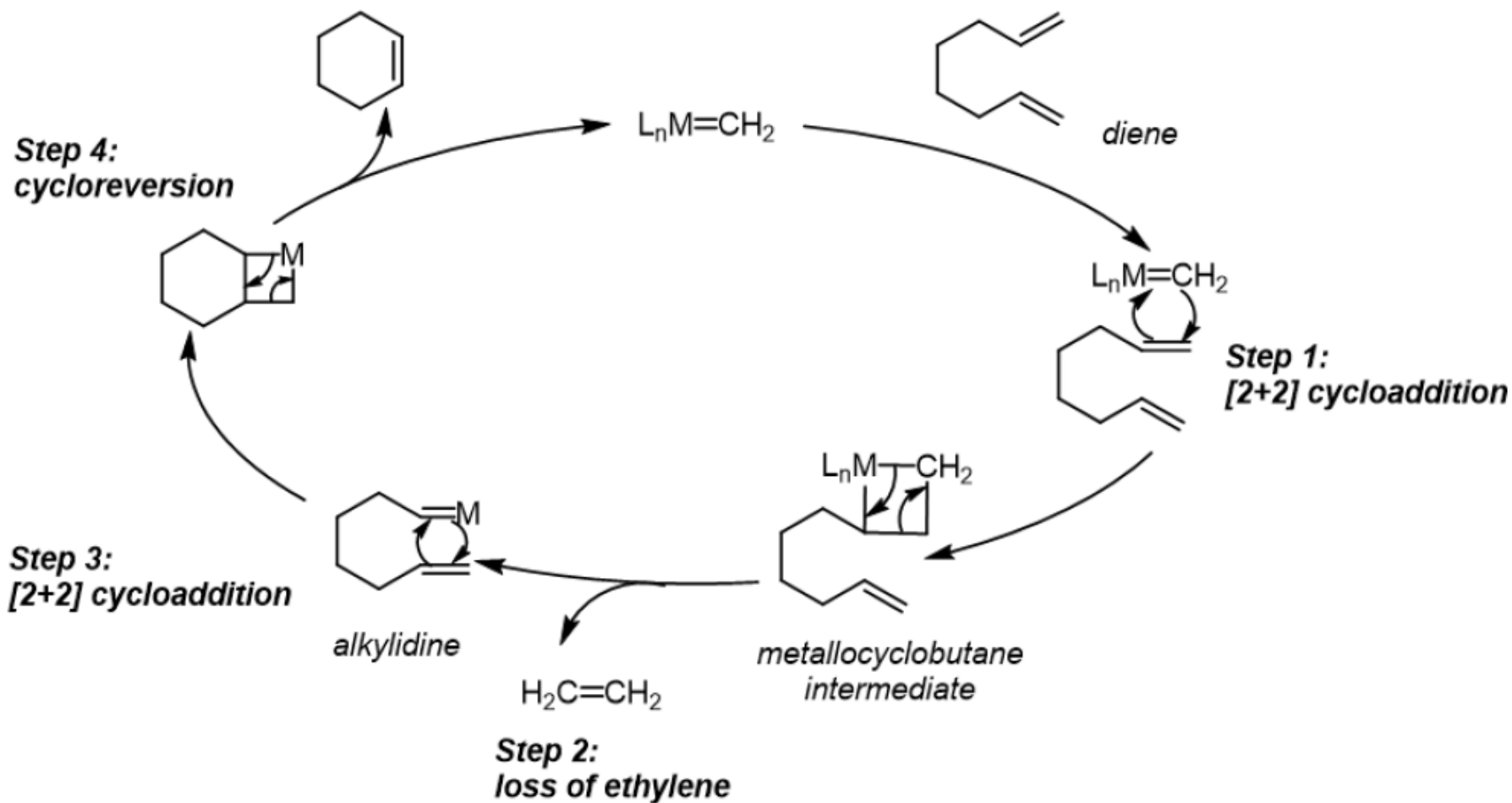
The Pauson–Khand reaction starts with the replacement of two CO molecules, one from each Co atom, with the alkyne to form a double σ -complex with two C–Co σ bonds, again one to each Co atom. One CO molecule is then replaced by the alkene and this π -complex in its turn gives a σ complex with one C–Co bond and one new C–C σ bond, and a C–Co bond is sacrificed in a ligand coupling reaction. Then a carbonyl insertion follows and reductive elimination gives the product, initially as a cobalt complex

Alkene (olefin) metathesis









Cross-Metathesis Reaction

step 1: First the ruthenium carbene catalyst undergoes [2 + 2] cycloaddition with the alkyne, then reverse [2 + 2] takes place and forms the new carbene complex.



step 2: Now the new carbene can undergo [2 + 2] cycloaddition and reverse [2 + 2] cycloaddition again, this time with the alkene component, and out comes the diene, plus a Ru carbene ready to start the cycle again.

